

Vimal Manoharan

+91 76038 32537 | 2004.vimal@gmail.com | linkedin.com/in/vimalmanoharan04 | github.com/vimal004

EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science Engineering (CGPA: 8.91/10.0)

Chennai, India

Sep. 2022 – May 2026

EXPERIENCE

Generative AI Trainee

Intel Unnati Program

Remote, India

Jan. 2024 – Mar. 2024

- Completed industrial training focused on Generative AI, LLM architectures, and agentic workflows.
- Architected data pipelines and deployment endpoints for LangChain-based AI language model agents.

React Native Frontend Developer Intern

Aakar Labs

Chennai, India

Sep. 2025 – Mar. 2026

- Collaborated within an Agile, product-focused team environment to design and pair-program iterative mobile interfaces for Aaku, an AI-powered travel companion utilizing React Native and Expo.
- Partnered closely with backend AI engineers to assist in the development and validation of an AI agent trip planning feature, ensuring dynamic frontend rendering of multi-agent execution workflows.
- Applied structured Object-Oriented patterns to engineer reusable UI components and state-driven flows across planning, trip progression, and contextual journey interaction modules.
- Integrated REST APIs and optimized frontend rendering performance, lowering end-user latency to support seamless, real-time client itinerary updates.
- Engaged in regular cross-functional feedback sessions with backend engineers and product designers to iteratively improve multi-step planning architectures and UX consistency.

Software Developer Intern

KSK Electronics Pvt. Ltd.

Chennai, India

Aug. 2025 – Oct. 2025

- Developed and refactored a full-stack ERP platform using React, Node.js, and MongoDB, migrating business-critical financial records from Zoho SaaS through clean, maintainable code architectures.
- Architected and deployed a corporate RAG (Retrieval-Augmented Generation) SOP framework using LangChain and Chroma DB, allowing employees to instantly retrieve exact policy explanations and FAQ answers.
- Implemented Object-Oriented design and role-based access control (RBAC), optimizing backend database queries to reduce stock retrieval latency.
- Established Continuous Integration (CI) concepts by containerizing services with Docker and deploying onto AWS EC2 with Nginx, significantly advancing delivery reliability and platform uptime.
- Followed test-focused logic to automate GST validation workflows, eliminating manual accounting and data-entry discrepancies by 90% during monthly audit operational cycles.

Freelance Full-Stack Developer

Siddha Shivalayas Clinic

Chennai, India

Jan. 2025 – Apr. 2025

- Listened thoughtfully to client business goals to design and iteratively build a MERN-stack Clinic Management System managing patient records, billing, and inventory workflows for 300+ active profiles.
- Adhered to SOLID principles to implement real-time inventory deduction logic, ensuring complete transactional database consistency during concurrent checkout operations.
- Built analytics dashboards using Material UI to map patient histories and financial trends, removing administrative bottlenecks and translating complex data into clear stakeholder value.

PROJECTS

Dental Clinic AI Receptionist Voice Agent | *FastAPI, RetellAI, Google Sheets API, SMTP, Render*

- Built a real-time AI voice agent managing inbound dental clinic calls for appointment booking, rescheduling, cancellations, and emergency handling via RetellAI Conversational Flow.
- Architected a dual-mode booking system featuring synchronous live slot availability checking via tool calls and asynchronous post-call background task execution.
- Engineered a FastAPI backend hosted on Render, integrating Webhook signature verification, Google Sheets API for automated data storage, and SMTP email notifications.

- Implemented non-blocking background tasks and a retry-once policy for critical integrations, preventing webhook timeouts and ensuring 99.9% processing reliability.

Explainable AI Framework for Early Risk Stratification | *Python, FastAPI, Scikit-learn, SHAP, NumPy*

- Designed and implemented an iterative predictive healthcare system utilizing ensemble machine learning models to run early clinical risk assessment and disease diagnosis.
- Engineered a probabilistic multimodal fusion pipeline to aggregate patient history and clinical metrics, translating unstructured medical data into predictive insights using clean analysis and coding principles.
- Integrated Explainable AI (XAI) frameworks utilizing SHAP and counterfactual analysis to extract feature importance, ensuring clinical transparent reasoning for diverse stakeholder perspectives.
- Built and deployed a low-latency FastAPI backend service structured to deliver real-time diagnostic predictions and visualization payloads securely.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML, CSS

Frameworks: React.js, React Native, Expo, Node.js, Express.js, Spring Boot, FastAPI, LangChain, LangGraph, RAG, Tailwind CSS, Material UI

Developer Tools: Git, Docker, AWS EC2, Nginx, Chroma DB (Vector DB), GitHub, Android Studio, VS Code

Methodologies & Libraries: AI Agents, RAG Pipelines, TDD, Agile, OOP, SOLID, SHAP, Pandas, NumPy, Scikit-learn, Selenium